PEOPLE FOR ANIMALS™

FIREWORKS
THE ANIMAL COST

New Year. Christmas. Weddings. Festivals. Public celebrations.
Measured consequences, explained mechanisms, targeted care.



THE BLAST ENDS BEFORE THE EFFECTS DO.

500 km
AN OBSERVED EXTREME

Some GPS-tracked geese made nonstop flights this long on fireworks night; this was not the average. [5]

42.9%
PERSISTING FEAR

Dogs reported fearful for over 30 minutes after hearing fireworks in a Dutch owner survey. [3]

1.5×
OXIDATIVE POTENTIAL

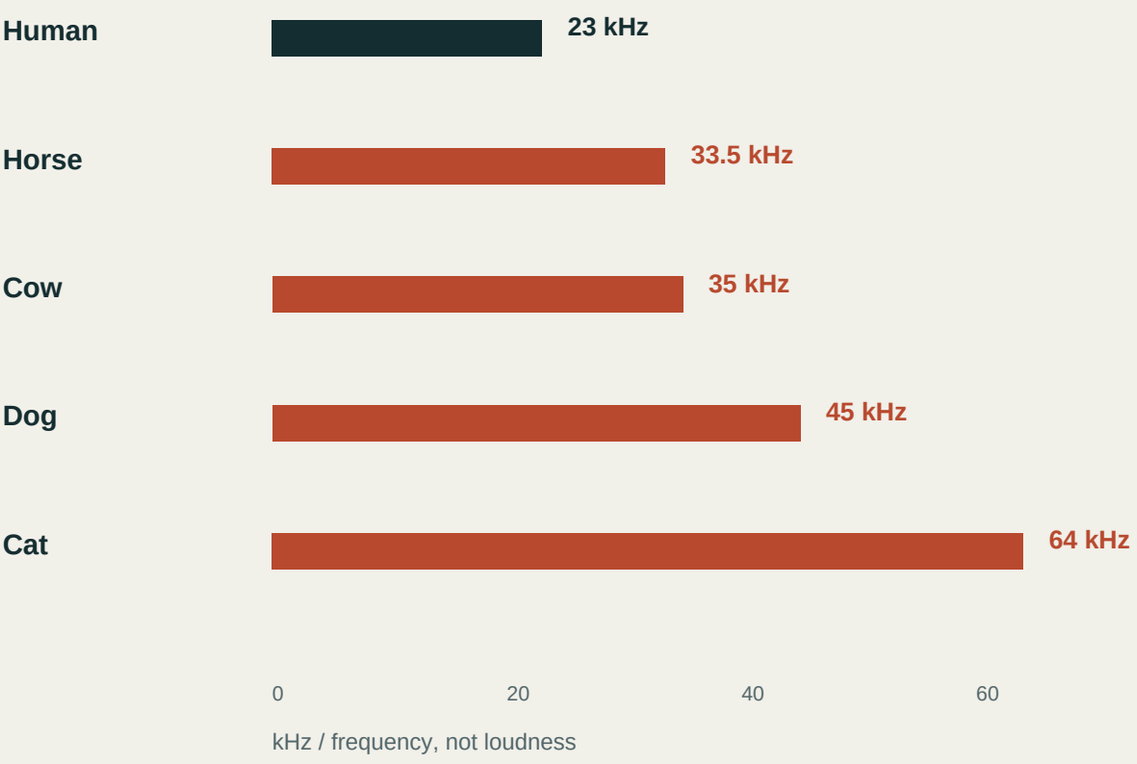
Laboratory assays of Mumbai particles rose by this factor in a study published in 2025. [8]

All photographic scenes are educational reconstructions; none is evidence of a measured event.

A different acoustic world.

Hearing frequency, sound intensity and fear are different quantities. A higher frequency limit does not mean a bang sounds a fixed number of times louder. [1,2]

APPROXIMATE UPPER AUDIBLE FREQUENCY



Approximate behavioural hearing limits compiled by Louisiana State University. Test methods and intensity criteria differ; age and individual hearing also matter. [1]

Damage can escape notice.

Noise trauma can cause acquired sensorineural hearing loss. In mammals, loss of cochlear sensory hair cells is irreversible. One-sided deafness can have few obvious signs. [2]

Hearing and fear can coexist.

BAER testing measures auditory pathway responses. It does not measure fear. A dog reacting less to sound after an exposure should not automatically be assumed to have become calmer. [2,12]

Care implication: prevent close blast exposure. Behaviour alone cannot rule out hearing injury, and human comfort with a sound is not a calibrated measure of an animal's exposure. [1,2]

The distress can outlast the noise.

A 2024 Dutch survey covered 3,009 dogs and 622 cats. The figures below are owner reports from a self-selected sample, not Indian prevalence or risks per display. [3]

FEAR LASTING MORE THAN HALF AN HOUR



8.4%
CATS EVER LOST

5.4%
DOGS EVER LOST

Reported lost at least once after hearing fireworks, including after fleeing. “Ever” is not a one-night probability. [3]

A quiet cat can still be afraid.

Freezing, hiding, appetite loss and withdrawal are included among firework-aversion behaviours. An absence of barking or running is not evidence of emotional recovery. [3]

Learned cues can spread fear.

A frightening noise can become associated with other cues, creating anxiety before the next blast. Controlled, low-intensity training differs from repeated exposure at a level that already causes fear. [12]

Use a recovery record: when did normal eating, movement and resting resume? Give your vet this timeline rather than only reporting that the animal “stopped reacting”. [12]

Owner-reported association is informative about welfare; it cannot diagnose each animal or prove causation.

The sky becomes an escape route.

Weather radar detects birds aloft even when people cannot see the nighttime disturbance. A Netherlands study combined radar with systematic bird counts. [4]



~1,000×

Average increase in birds in flight on New Year's Eve compared with other nights. Large-bodied bird communities reacted more strongly. [4]

~10 km

DISTURBANCE STILL DETECTABLE

Flight activity remained roughly tenfold higher at this distance from residential areas. [4]

A boundary is not a shield.

The response declined most steeply within the first 5 km. These data do not establish a safe firing distance: displays, habitats and bird communities differ. [4]

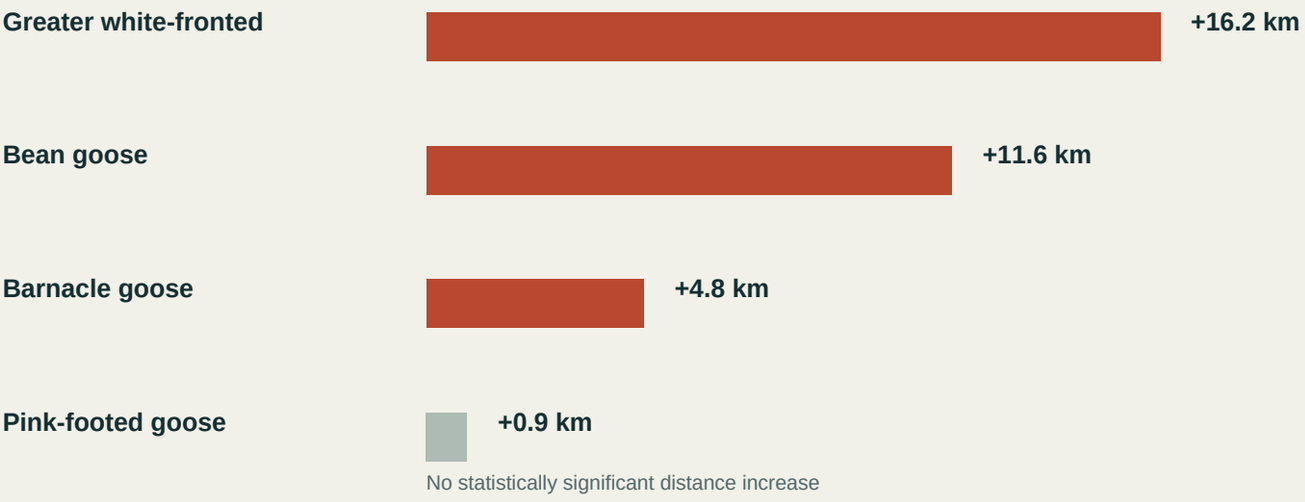
Ecological implication: limiting the launch point alone may leave distant roosts exposed. Keep firework events away from concentrations of resting birds and protect quiet refuges across the surrounding area. [4,16]

A few minutes can alter the night.

GPS tracking followed 347 migratory geese of four species across eight New Year periods in Germany, Denmark and the Netherlands. [5]

CHANGE IN NIGHTTIME MOVEMENT EXTENT

Model-estimated increase relative to preceding nights; species means. [5]



40–150 m

HIGHER NIGHT FLIGHTS

All four species increased maximum flight height on average. [5]

1.9 h

LESS ROOST TIME

Barnacle geese lost about this much nightly roost time; the result did not apply equally to all species. [5]

500 km

AN EXTREME, NOT A MEAN

Some recorded movements reached this scale. Most species' mean distance increases were far smaller. [5]

The delayed cost: most geese increased feeding in the days afterwards, consistent with compensation for extra flight expenditure. The study tracked the 12 days after the event. [5]

A separate paper published in 2026 found that disturbance behaviours and gull roost numbers took several days to recover. Its observations spanned winters 2020-2023. [6]

MOVEMENT EXTENT IS NOT TOTAL FLIGHT PATH LENGTH / VALUES ROUNDED

A concentration is only half the story.

Particle mass describes exposure; chemical reactivity describes another part of the hazard. Neither alone predicts an individual animal's clinical outcome. [8,11]

DELHI / 9 NOVEMBER 2026 / 02:00 IST

3,744

µg/m³ PM2.5 at Mundka.
11 stations recorded over 2,000 µg/m³ in DPCC data reported the same day. [7]

Hourly readings: not citywide means, AQI values or fireworks-only fractions.

MUMBAI / RESIDENTIAL PARTICLES / STUDY PUBLISHED 2025

~2.6×

PM2.5 MASS

Event-night concentration relative to a comparison night. [8]

~3×

SELECTED METALS

Zn, Pb, Mn and Cu concentrations increased. [8]

~1.5×

OXIDATIVE POTENTIAL

Volume-normalised DTT and AA assays increased. [8]

Five-day residential monitoring with chemical and laboratory redox assays. These multipliers describe different endpoints and must not be treated as interchangeable "toxicity scores". [8]

What the assays add

DTT and ascorbic-acid assays estimate the capacity of sampled particles to drive oxidation or deplete antioxidants. This is laboratory evidence of reactive chemistry, not a measured increase in animal deaths. [8]

Why animal biology matters

Inhaled particles affect different airway regions according to size and chemistry. Birds have greater ventilation and respiratory surface area per unit mass, increasing susceptibility to inhaled toxicants. [11]

Spent does not mean chemically inert.

A 2026 laboratory study examined how firecracker residues changed dissolved organic matter in river, lake and ultrapure water. [9]



01

RELEASE

Residues released dissolved organic compounds, including low-molecular-weight and sulfur-containing material. [9]

02

ADSORPTION

Other dissolved compounds attached to the solid residues; water chemistry changed in both directions. [9]

03

WATER DEPENDENCE

Initial water chemistry altered the result: the same residue did not produce a universal response. [9]

What this establishes

A route for post-event chemical exposure in aquatic habitats. It does not establish a field mortality rate for fish, a safe residue dose, or the effect in every Indian waterbody. [9]

What changes the outcome

Collect cooled residues before runoff carries them to drains or waterbodies. Do not wash celebration litter into a gutter. This interrupts the exposure route the study investigated. [9,16]

Each barrier has a specific job.

A refuge reduces exposure; it does not erase a learned fear. Introduce it before the event and preserve the animal’s ability to choose its hiding place. [12]



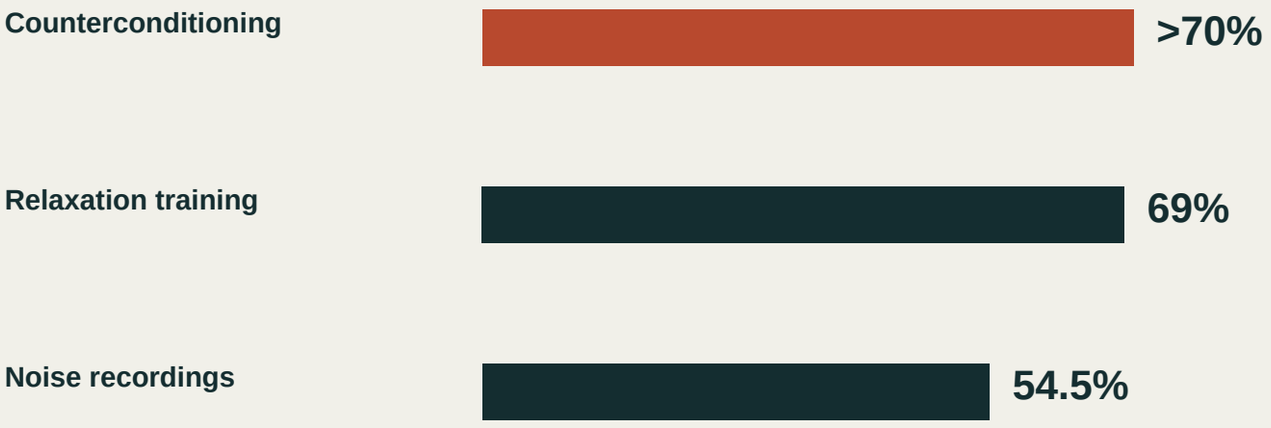
BARRIER	WHAT IT ADDRESSES	WHAT IT DOES NOT PROVE
1 Curtains	Reduce visible flashes.	Ordinary curtains do not filter outdoor smoke.
2 Open refuge	Permits voluntary withdrawal.	A covered crate is not soundproof or automatically reassuring.
3 Secured exit	Reduces escape opportunities.	Preventing escape does not itself treat fear.

Airflow is a separate requirement: preserve safe temperature and ventilation. A fan circulates air; it does not remove particles. A dark, hot or smoke-filled enclosure is not a protective refuge. [11,12]

Quiet behaviour is not a treatment test.

Owner-perceived success in a 1,225-response study, summarised in a 2023 evidence review. These were not randomly assigned treatment groups. [13,14]

REPORTED SUCCESS / DIFFERENT USERS AND DENOMINATORS



This comparison cannot establish which approach is superior for a given dog. Owner expectations, fear severity, treatment selection and implementation can affect reported outcomes. [13,14]

Change the association

Counterconditioning pairs the sound with something the dog values. Offer food or play if the dog is willing; forced feeding is not the technique. Refusal can signal that stress is too high for that approach at that moment. [14]

Keep exposure below fear

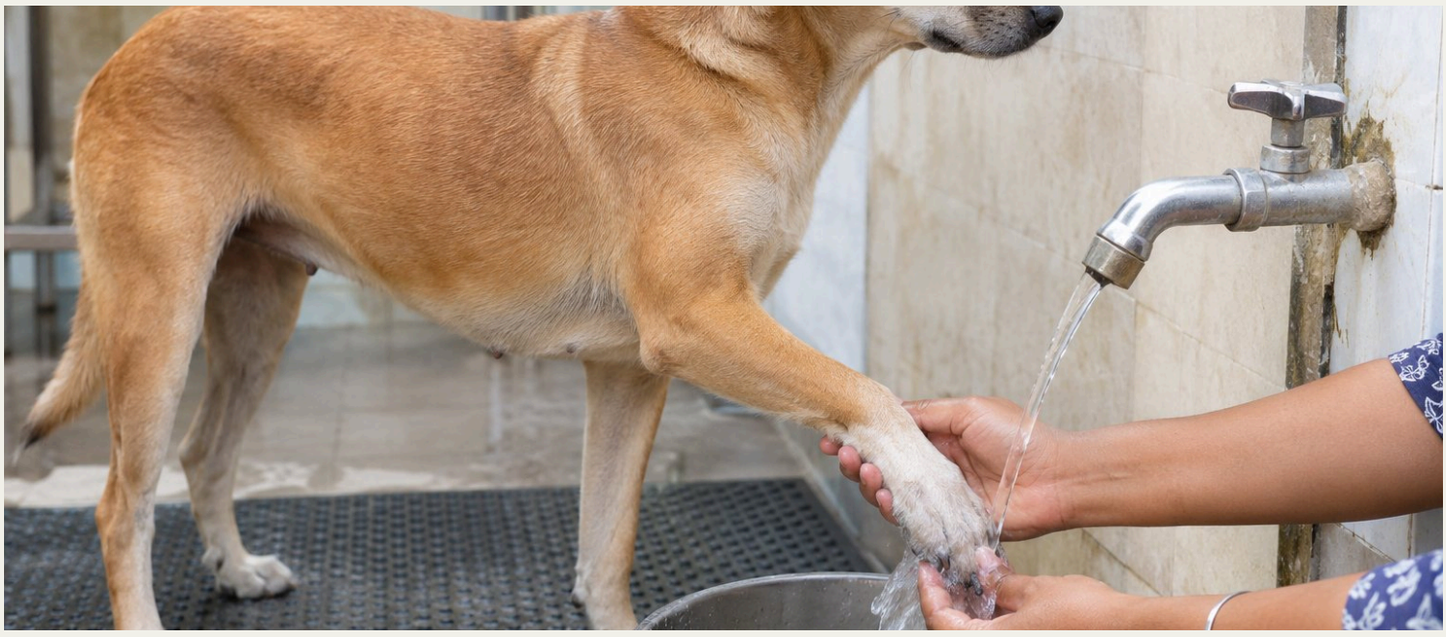
Desensitisation begins at an intensity that does not elicit anxiety. A recording may omit high frequencies, flashes, odours and vibration; tolerating playback does not demonstrate tolerance of live fireworks. [12,14]

Medication targets anxiety, not just movement. Acepromazine alone is unsuitable for noise fear because tranquillisation does not provide reliable anxiety relief. Ask a vet for an individual plan and trial it before the event; no self-selected doses. [14]

Environmental protection remains necessary while treatment is underway; training is not a reason to expose an animal to a live display.

Act on the pathway of injury.

Noise fear, thermal burns, chemical ingestion and smoke injury can coexist. Each requires a different response; apparent calm cannot exclude physical harm. [10,11,15]



20 min

Cool a thermal skin burn with clean running water promptly. RECOVER recommends 20 minutes; benefit is supported when cooling starts within 3 hours. Keep the rest of the body warm. [15]

Cool water, not ice. Do not pour water onto the face or risk chilling the whole animal. The illustrated paw position shows cooling, not a substitute for veterinary examination. [15]

Swallowed firework

Unused pyrotechnics may injure the gut, kidneys, liver or bone marrow, or provoke seizures. Effects depend on contents. Contact a vet; take packaging and do not induce vomiting unless instructed. [10,15]

Delayed airway injury

After substantial smoke exposure, respiratory injury may develop over 12-24 hours or later despite few external signs. Trouble breathing, collapse or altered consciousness warrants emergency veterinary assessment. [11]

For community animals: contact a vet or local PFA unit and describe exposure, timing and signs. Immediate danger or deliberate cruelty in India: 112. Keep a clear route for professional rescue rather than surrounding a frightened animal. [16]

What each source can tell us.

Tap a source title to open it. Study location and observation dates matter; publication year alone does not make the underlying measurements recent.

01

Louisiana State University

How Well Do Dogs and Other Animals Hear?

Comparative hearing limits. Approximate values depend on test criteria; they are not loudness multipliers.

02

Merck Veterinary Manual, updated 2026

Deafness in Animals

Clinical review: noise trauma, sensorineural loss and BAER. Mechanisms are not fireworks-specific injury rates.

03

Van Herwijnen et al., 2024

Firework aversion in cats and dogs as reported by Dutch animal owners

Veterinary and Animal Science 26:100402. 3,009 dogs and 622 cats; self-selected reports collected in 2023.

04

Hoekstra et al., 2024

Fireworks disturbance across bird communities

Frontiers in Ecology and the Environment 22:e2694. Netherlands radar and bird counts; disturbance, not mortality.

05

Kölzsch et al., 2023; online 2022

Wild goose chase: geese flee high and far, and with aftereffects

Conservation Letters 16:e12927. 347 geese, four species; GPS from 2014-2021 in continental Europe.

06

Moser, 2026

Birds on fire: severe disturbance by fireworks in wintering birds

Animal Behaviour 239:123670. Published 2026; winter observations 2020-2023. Recovery of roost size and behaviour.

07

DPCC readings reported by The New Indian Express, 9 Nov 2026

Delhi festival-night PM2.5 data released

Secondary reporting of hourly official station readings. No estimate here attributes the entire concentration to fireworks.

08

Dubey et al., 2025

Chemical and toxicological characteristics of fine particles from festive fireworks

Atmospheric Environment 350:121160. Mumbai residential monitoring and oxidative-potential assays; not animal outcomes.

Sources checked 10 November 2026. Educational reconstructions are AI-generated and do not document measured events. PFA logo supplied by requester; independently prepared. Clinical examination takes precedence over this guide.

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09 **Chen et al., 2026**

Molecular-level perturbations of dissolved organic matter by residue leaching

Environmental Science & Technology 60:16168-16178. Laboratory river/lake-water chemistry; no field fish mortality measured.

10 **Pet Poison Helpline**

Fireworks: toxicity to pets

Veterinary toxicology summary: contact burns and ingestion. Formulation, dose and animal determine severity.

11 **Merck Veterinary Manual, updated 2026**

Smoke Inhalation Injury in Animals

Clinical toxicology: susceptibility, airway injury and delayed progression. Severe smoke exposure differs from ambient haze.

12 **Merck Veterinary Manual**

Behavior Problems of Dogs

Clinical guidance on fear learning, graded exposure and a previously familiar safe haven.

13 **Riemer, 2020**

Effectiveness of treatments for firework fears in dogs

Journal of Veterinary Behavior 37:61-70. Owner-perceived effectiveness; observational treatment comparisons.

14 **Riemer, 2023**

Therapy and prevention of noise fears in dogs: evidence review

Animals 13:3664. Management, training and anti-anxiety treatment; clinical choices require individual assessment.

15 **Mandell et al., 2026**

RECOVER Guidelines: First Aid for Dogs and Cats

Journal of Veterinary Emergency and Critical Care. Evidence and consensus guidelines for prehospital first aid.

16 **People for Animals, India**

Fireworks and animals; report cruelty

India-focused care and reporting. Injured animals need veterinary care first; reporting: peopleforanimalsindia.org/report.

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